YOHANNES BELAI

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PROFESSIONAL SUMMARY

Computer Science student graduating May 2027 with hands-on experience building full-stack web applications and evaluating software implementations. Strong foundation in **Java, Python, JavaScript, TypeScript, SQL, data structures and algorithms, APIs, databases, and software engineering**. Experienced in debugging, code review, AI API integration, authentication, and communicating technical findings clearly.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL

Web & Backend: React, Vite, Tailwind CSS, Node.js, Express.js, Spring Boot, REST APIs, HTML/CSS

Databases: PostgreSQL, MongoDB (schema design, indexing), SQLite

Authentication & Security: OAuth 2.0, JWT, bcrypt, Role-Based Access Control (RBAC)

AI & Data Processing: Google Gemini API, JSON, structured data extraction, Apache POI, prompt design

Engineering & Tools: Data Structures & Algorithms, OOP, Debugging, Code Review, Git, GitHub, VS Code, Emacs

EDUCATION

Towson University — Towson, MD | Bachelor of Science in Computer Science | Expected May 2027

Relevant Coursework: Database Management Systems, Data Communications & Networking, Operating Systems, Ethical & Societal Computing

Montgomery College — Associate of Science in Computer Science

SOFTWARE DEVELOPMENT EXPERIENCE

Alignerr — Software Developer (Contract) | Remote | Aug 2025 – Present

- Evaluate AI-generated Java and Python solutions for algorithmic correctness, logical flow, edge cases, task requirements, and software engineering principles.
- Conduct structured code reviews and debugging to identify logical defects, implementation issues, edge-case failures, and potential security vulnerabilities.
- Research implementation issues and produce detailed technical evaluations and actionable feedback supporting AI training and evaluation datasets.
- Apply data structures, algorithms, object-oriented programming, and clean-code principles across varied technical requirements.

Tech Education Programs — Primary Teacher & Teaching Assistant (Intern) | Jun 2026 – Aug 2026

- Taught Python programming, object-oriented programming, and algorithm design to 20+ students using MIT App Inventor.
- Reviewed student software projects, diagnosed implementation issues, and resolved technical blockers through structured debugging and code review.
- Communicated technical concepts and debugging approaches clearly to students with varying levels of programming experience.

PROJECTS

Drill Live — Full-Stack AI Study Platform | Spring Boot, React, Java, REST APIs, Google Gemini API

- Engineered a full-stack application that processes multi-file PowerPoint uploads and extracts presentation content using Apache POI.
- Integrated the Google Gemini API to transform extracted text into structured JSON flashcards and quizzes through API workflows and prompt design.
- Built an end-to-end workflow connecting a React frontend, Spring Boot backend, document-processing logic, and external AI service.

SyllabiXtract — React Web Application | React, TypeScript, Vite, Tailwind CSS, OAuth 2.0, REST APIs

- Developed a React interface supporting multi-file asynchronous uploads and real-time progress feedback for document-processing workflows.
- Implemented OAuth 2.0 authentication through Auth0 and connected authenticated frontend workflows to REST APIs.

Campus Events API — Backend Application | Node.js, Express.js, MongoDB, JWT

- Designed and implemented 12 RESTful API endpoints for event scheduling and user registration using Node.js and Express.js.
- Implemented JWT authentication, bcrypt password hashing, and role-based access control for administrator, organizer, and user permissions.
- Designed MongoDB schemas and database indexes to support queries involving event dates and user status.